

Peripheral Veno-Arterial-Extracorporeal Membrane Oxygenation for Refractory Septic Shock in Children: A Multicenter Review

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Abstract

Background: Extracorporeal membrane oxygenation (ECMO) is utilized as a rescue therapy in the management of pediatric patients with refractory septic shock. Multiple studies support the use of a central cannulation strategy in these patients. This study aimed to assess the survival of and identify mortality risk factors in pediatric patients supported with peripheral veno-arterial (VA) ECMO in the setting of septic shock. **Methods:** We retrospectively reviewed and compared clinical characteristics of 40 pediatric patients supported with peripheral VA ECMO for refractory septic shock, at two tertiary care children's hospitals from 2006 to 2020. Our hypothesis was that peripheral VA ECMO is effective in supporting cardiac function and improving tissue oxygenation in most pediatric patients with refractory septic shock. **Results:** The overall rate of survival to discharge was 52.5%, comparable to previously reported survival for pediatric sepsis on ECMO. With the exclusion of patients with an oncologic process, the survival rate rose to 62.5%. There was a statistically significant difference in mean pump flow rates within 2 hours of initiation of ECMO between survivors and non-survivors (98 mL/kg/min vs 76 mL/kg/min, $P = .050$). There was no significant difference between pre-ECMO vasoactive inotropic score (VIS) in survivors and non-survivors. A faster decrease in VIS in the first 24 hours was associated with lower mortality. **Conclusions:** From this large case series, we conclude that peripheral VA ECMO is a safe and effective modality to support pediatric patients with refractory septic shock, provided there is establishment of high ECMO pump flows in the first few hours after cannulation and improvement in the VIS.

Keywords

ECMO, pediatric, septic shock, mortality, peripheral, VA ECMO

Introduction

Sepsis is the leading cause of morbidity and mortality in children worldwide, with more than 6800 pediatric sepsis-related deaths in the United States annually.¹ The single most significant risk factor for mortality in sepsis is the development of refractory shock.^{2,3} The American College of Critical Care Medicine (ACCM) defines refractory septic shock as a shock that persists despite goal-directed use of inotropes, vasopressors, vasodilators, and maintenance of metabolic and hormonal homeostasis.⁴ The Surviving Sepsis Guidelines and the ACCM recommend extracorporeal membrane oxygenation (ECMO) as a potentially life-saving treatment modality for pediatric patients with refractory septic shock.⁴⁻⁶

ECMO is a cardiopulmonary bypass modality that supports the failing heart and/or lungs of critically ill children. It serves as a bridge therapy until the return of native function or until transplantation. There are two techniques for cannula placement in ECMO. One approach is peripheral, utilizing the femoral artery or the carotid and the femoral vein or the internal jugular vein. The other is central, directly through the right atrium and aorta.⁷ The theoretical benefit of central cannulation

includes achieving higher ECMO flow rates, potentially reversing shock and multiorgan dysfunction more rapidly than might be accomplished by other cannulation strategies.⁸ Previous

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studies indicated central ECMO, with direct cardiac cannulation, is superior for septic patients.^{9–12} However, central cannulation is a high risk, invasive procedure which independently increases the risk of bleeding on ECMO.¹³ Cardiac surgeons, a limited resource, are needed for central cannulation. Conversely, peripheral ECMO can be done percutaneously and at the bedside. In some institutions, this is performed by intensivists.⁹ Furthermore, circuit flows of >100 mL/kg/min and up to 150 mL/kg/min can routinely be achieved with peripheral cannulation techniques, potentially negating the theoretical benefits of central cannulation.

We hypothesize that peripheral veno-arterial (VA) ECMO is effective in supporting cardiac function and improving tissue oxygenation in most pediatric patients with refractory septic shock.

Using data from two tertiary care children's hospitals, we reviewed patients supported with peripheral VA ECMO cannulation through cervical or femoral vessels for refractory septic shock. Both these institutions have routinely used peripheral VA ECMO cannulation as an effective and safe method of supporting these patients in a less invasive manner. The aims of this study were to assess survival and identify risk factors for mortality in pediatric patients supported with peripheral VA ECMO in the setting of septic shock.

Methods

Study Design and Patient Characteristics

In this multicenter retrospective study, all patients between 30 days and 18 years of age treated with ECMO for septic shock

according to the International Consensus Conference on Pediatric Sepsis were eligible for inclusion. This definition of septic shock in children requires two or more of the systemic inflammatory syndrome criteria, suspected or proven infection, and the need for vasoactive support to reach a mean arterial blood pressure (age targeted) despite adequate fluid resuscitation.

Data were collected from 2006 to 2020 at Riley Hospital for Children at Indiana University Health in Indianapolis, Indiana (IU) and from 2010 to 2017 at Children's Health/UT Southwestern Medical Center in Dallas, Texas (UTSW). We collected patient demographics pre-ECMO, on-ECMO hemodynamic data points, mechanical ventilator support data, cannulation sites, lab values, mortality, and complications. We also calculated vasoactive inotropic scores (VIS), a weighted sum of all administered inotropes and vasoconstrictors that reflects the pharmacological support of the cardiovascular system. We looked at outcomes including mortality, duration on ECMO, length of stay in the ICU and length of hospitalization at the time of discharge from the ICU (intensive care unit).

Figure 1 illustrates our included cohort. Patients who were placed on ECMO for diagnoses unrelated to sepsis were excluded. Other exclusions included patients with multiple ECMO runs, centrally cannulated patients, patients requiring conversion from VV (veno-venous) to VA, and any patients supported on VV ECMO. No patients required ECPR in this cohort.

IRB approval was obtained at both institutions and a data sharing agreement was established between UTSW and IU. The data was collated, and statistical analysis was performed at IU.

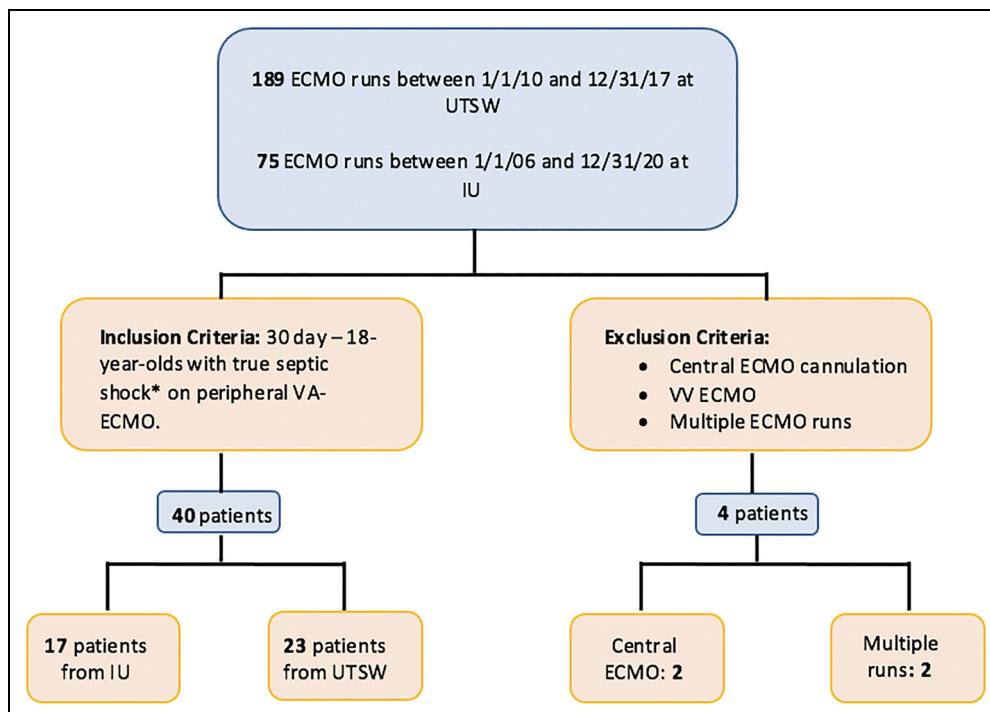


Figure 1. Illustrates our patient cohort.

Statistical Analysis

Simple regression models were fit to investigate the effect of demographic factors, infection source, lab results, VIS scores, ventricular dysfunction, pump flow rates on mortality and prevalence of complications on ECMO specifically renal and CNS complications and duration on ECMO.

Logistic models were fit for outcomes of mortality, prevalence of complications on ECMO, prevalence of renal failure, and prevalence of CNS complications. A simple logistic regression model was fit between mortality odds and pre-to-post initiation of ECMO changes in VIS, and the model was used to obtain predicted probabilities of mortality, based off a 50% cutoff in mortality, for each patient, with sensitivity and specificity reported. A negative binomial model was fit for time on ECMO.

All regression analyses were fit using the *brms* R package, version 2.16.¹⁴ We defined a significance as equal or less than $P = .050$.

Results

Demographics

On review of the records at both IU and UTSW, 264 pediatric patients without congenital cardiac disease were supported with ECMO. As illustrated in Figure 1, 40 patients with septic shock underwent peripheral cannulation for ECMO and were included in our analysis. Of these, 17 (42.5%) patients were from IU and 23 (57.5%) were from UTSW. No patients required E-CPR cannulation in our cohort.

Table 1 shows the demographics and baseline patient characteristics of patients from both institutions. The median age at ECMO initiation was 88 months (range: 1 month–16.8 years). The median body weight was 25 kg (range: 3–100.1 kg). The median time on ECMO was 4 days (range: 0–53 days). The mean max pump flow rate within the first 2 hours of cannulation was 90 mL/kg/min. Mortality occurred in 19 (47.5%) patients. In immunosuppressed oncology patients, mortality occurred in 6/10 patients (60%) versus 13/30 (37.5%) non-oncology patients.

Echocardiographic evaluation was available in 27 (69%) of the cases. Reduced single- or bi-ventricular cardiac function was reported in 14 (52%) patients. The most commonly used vasoactive agent was epinephrine, with a median number of three vasoactive medications per patient.

Positive microbiological cultures were identified in 32 (80%) patients, with bacterial pathogens predominating in 26 (65%).

Mortality and Risk Factors

Twenty-five patients survived to ECMO decannulation (62.5% survival), with 21 surviving to ICU discharge (52.5% survival). The primary cause of death was multiorgan failure. In all cases of death on ECMO, ECMO support was discontinued due to futility.

Table 2 illustrates that increased mortality was associated with increasing age at ECMO initiation (OR 2.63, $P < .05$), increasing admission weight (OR 2.21, $P < .01$) and decreasing pre-ECMO platelet count (OR 0.4, $P < .05$). Patients older than 51 months (4.25 years) had a greater likelihood of mortality. This was also the case with patients heavier than 26.8 kg. We also found that patients with pre-ECMO platelet count lower than 30,000 platelets had an increased likelihood of mortality.

Vasoactive Inotropic Scores

VIS were calculated as illustrated in Figure 2.¹⁵ VIS was utilized as a measure of hemodynamic dysfunction. Pre-ECMO VIS was calculated utilizing the highest dosages of supportive drips during the 24 hours prior to cannulation. Change in VIS on ECMO explored the change of inotropic support over the first 24 hours of ECMO. Both factors were analyzed as potential predictors of mortality.

There was no difference between pre-ECMO VIS in survivors versus non-survivors ([95%CI] = 0.004[−2.74, 2.71]), and pre-ECMO VIS had no significant association with mortality odds (OR [95%CI]: 1 [0.55, 4.57]). A model based on a 50% predicted mortality was used to predict mortality based off change in VIS score over the first 24 hours. This model showed that an increase in VIS was predictive of higher mortality, while a decrease in VIS was more likely to result in survival. Figure 3 demonstrates this model.

ECMO Run Data

The mean max pump flow rate within 2 hours of initiation was 90 mL/kg/min (range: 59.8–112.5). We found that there is a statistically significant difference in mean pump flow rates within the first 2 hours between survivors and non-survivors (98 mL/kg/min vs 76 mL/kg/min, $P = .050$).

The median time on ECMO was 4 days (range: 0–53). Variables shown in Table 3 were analyzed to determine the correlation with the duration of ECMO. Table 3 demonstrates the incidence rate ratios and 95%CI for potential risk factors for the duration of ECMO support. We did not find any correlation between age, weight, immunosuppression, pH, platelet count, and diastolic dysfunction with the duration of ECMO support. Patients with viral septic shock were found to require longer ECMO support. Higher pump flow rates within 2 hours of initiation of ECMO support were associated with decreased time on ECMO, with every change in pump flow rate by 1 Z score (1 SD) leading to an estimated decrease in time on ECMO by 0.5 days.

Complications

There was an overall complication rate of 57.5% ($n = 23$). Acute kidney injury (AKI) was the most common complication, noted in 18 (45%) of the patients. Seventeen patients (37.7%) had a bleeding complication, with 8 (20%) having an intracranial bleed.

Table 1. Clinical Characteristics of Patients with Septic Shock Treated with ECMO (n = 40).

Variable	All patients, Median [I.Q, 3Q] or Count (%)	All patients, Median [IQR] or Count (%)	All patients, Mean [Min, Max]	All patients, Range	Riley (n = 17), Median [I.Q, 3Q] or Count (%)	UTSW (n = 23), Median [I.Q, 3Q] or Count (%)	P-Value
Institution							
Riley	17 (42.5)	17 (42.5)				9 (39.1)	1.000
UTSW	23 (57.5)	23 (57.5)				90 [32, 159]	.632
Female gender	15 (37.5)	15 (37.5)			6 (35.3)	25 [14, 50.2]	.870
Age at ECMO (months)	88 [30.8, 145]	88 [11.4, 2]	89.7 [0, 202]	202	86 [14, 141]	29.5 [15.5, 41]	.223
Weight at admission (kg)	25 [12.4, 44.9]	25 [32.5]	32.6 [3, 100.1]	97.1	26.8 [11.7, 40.5]	3 [0, 16.8]	.015
VIS Pre-ECMO	31.5 [19.4, 41.5]	31.5 [22.1]			38 [29.4, 41.4]	-16 [-32.4, -9]	.711
VIS Post-ECMO	8.2 [1.9, 21.7]	8.2 [19.8]			14 [7.5, 28]	5 (21.7)	.444
VIS change, Post - Pre	-15 [-27.8, -7.2]	-15 [20.6]			-14.3 [-21.9, -4]	4 (17.4)	.521
Oncology patient	10 (25.6)	10 (25.6)			5 (31.2)	2 (8.7)	.294
Immunosuppressed	9 (23.1)	9 (23.1)			5 (31.2)	10 (43.5)	.022
Bacterial growth	26 (65)	26 (65)			10 (58.8)	11 (47.8)	.755
Fungal growth	2 (5)	2 (5)			0 (0)	4 (17.4)	.373
Viral growth	4 (10)	4 (10)			3 (17.6)	7.2 [7.1, 7.2]	.132
Blood infection	24 (60)	24 (60)			14 (82.4)	62.2 [49.4, 164.7]	.015
Lung infection	18 (45)	18 (45)			7 (41.2)	2 [1.6, 3.4]	.501
Urine infection	5 (12.5)	5 (12.5)			1 (5.9)	78 [39, 128.5]	.511
Worst pH pre-ECMO	7.1 [7, 7.2]	7.1 [0.2]			7.1 [7, 7.2]	-10 [-15, -5]	.038
PTT pre-ECMO	53.5 [42.1, 91.9]	53.5 [49.8]			41.8 [36.9, 46.5]	15 (65.2)	.726
INR pre-ECMO	1.9 [1.6, 2.8]	1.9 [1.2]			1.8 [1.6, 2.3]	91 [73.6, 120.7]	.337
Platelets pre-ECMO	85 [42, 145]	85 [103]			93.5 [48.2, 157.8]	3 [2, 3]	.034
BDEF pre-ECMO	-14 [-18, -6.5]	-14 [11.5]			-16.5 [-19, -13]	12 (52.7)	<.001
Pre-ECMO ECHO problems	14 (51.9)	14 (51.9)			12 (75)		
Pump flow rate, 2 hours	90 [59.8, 112.5]	90 [52.7]	87.5 [12, 155]	143	82.8 [54.6, 99.6]		
Vasopressor number	3 [2, 3]	3 [1]			3 [2, 4]		
Vasopressor used							
Vasopressin	20 (50)	20 (50)			8 (47.1)	12 (52.2)	
Norepinephrine	12 (30)	12 (30)			12 (70.6)	0 (0)	
Epinephrine	36 (90)	36 (90)			16 (94.1)	20 (87)	
Dopamine	16 (40)	16 (40)			11 (64.7)	5 (21.7)	
Dobutamine	4 (10)	4 (10)			4 (23.5)	0 (0)	
Milrinone	13 (32.5)	13 (32.5)			12 (70.6)	1 (4.3)	
Mortality	19 (47.5)	19 (47.5)			9 (52.9)	10 (43.5)	.750
Any ECMO complication	23 (57.5)	23 (57.5)			15 (88.2)	8 (34.8)	.001
CNS complications (bleeding and/or clotting)	7 (17.5)	7 (17.5)			5 (29.4)	2 (8.7)	.113
Renal failure	18 (45)	18 (45)			14 (82.4)	4 (17.4)	<.001
Time on ECMO (days)	4 [2, 6.2]	4 [4.2]	6 [0, 53]	53	5.8 [3.2, 10.9]	3.9 [1.7, 5.2]	.031
Time on vent (days)	9 [3, 15]	9 [12]	12.7 [0, 67]	67	12 [7, 21]	8 [2, 12]	.037

Abbreviations: ECMO, extracorporeal membrane oxygenation; VIS, vasoactive inotropic score; PTT, partial thromboplastin time; INR, international normalized ratio; ECHO, echocardiogram.

Table 2. Predictors of Mortality.

Independent variable	Odds ratio [95%CI]	P-Value
Age at ECMO *	2.63 [1.3, 5.38]	.008
Admission weight *	2.21 [1.07, 4.57]	.032
VIS *	1 [0.55, 1.82]	.899
Oncology patient	3.31 [0.78, 13.98]	.103
Bacterial growth	0.58 [0.16, 2.01]	.393
Fungal growth	1.08 [0.1, 11.68]	.951
Viral growth	0.42 [0.06, 3.16]	.398
Blood infection	2.8 [0.79, 9.88]	.111
Lung infection	0.56 [0.17, 1.89]	.345
Urine infection	0.31 [0.05, 2.18]	.239
Worst pH pre-ECMO *	0.61 [0.31, 1.17]	.584
PTT pre-ECMO *	0.94 [0.5, 1.78]	.854
INR pre-ECMO *	1.13 [0.6, 2.11]	.713
Platelets pre-ECMO *	0.43 [0.19, 0.97]	.042
Pre-ECMO ECHO problems	1.22 [0.27, 5.47]	.799
Pump flow rate, 2 hours *	0.67 [0.35, 1.29]	.236
Vasopressor Number	1.18 [0.59, 2.35]	.641

Notes: Variables with a * after them have been converted to Z-scores; they have been scaled so that their mean is 0, and their standard deviation is 1.

Bold denotes statistical significance.

Abbreviations: ECMO, extracorporeal membrane oxygenation; VIS, vasoactive inotropic score; PTT, partial thromboplastin time; INR, international normalized ratio; ECHO, echocardiogram.

$$\text{VIS} = \text{dopamine dose } (\mu\text{g/kg/min}) + \text{dobutamine dose } (\mu\text{g/kg/min}) + 100 \times \text{epinephrine dose } (\mu\text{g/kg/min}) + 10 \times \text{milrinone dose } (\mu\text{g/kg/min}) + 10,000 \times \text{vasopressin dose } (\text{U/kg/min}) + 100 \times \text{norepinephrine dose } (\mu\text{g/kg/min})$$

Figure 2. Calculation of VIS.

Discussion

This is a retrospective cohort study of pediatric patients with refractory septic shock supported with peripheral VA ECMO. The study focused on patients receiving peripheral VA ECMO support only, capturing the most critically ill, non-ECPR patients with septic shock at both institutions. The overall survival rate to discharge was 52.5% and rises to 62.5% when patients with an oncologic process were excluded. This is comparable to the ELSO registry and other studies that include central cannulation.^{13,16} Our results indicate that peripheral VA ECMO can adequately support most children with septic shock.

Our results suggest that higher ECMO flow rates early in treatment are associated with better survival. The mean pump flow in survivors, within 2 hours of initiation was 98 mL/kg/min, compared to 76 mL/kg/min in non-survivors ($P = .050$). Additionally, a larger decrease in VIS in the first 24 hours was also associated with lower mortality despite no significant difference between pre-ECMO VIS in survivors and non-survivors. A decrease in VIS and timely restoration of tissue

perfusion by adequate ECMO flows indicate that these patients are being optimally supported, a well-described prognostic indicator.^{17,18}

Patients over 51 months of age and those weighing more than 26.8 kg were less likely to survive to decannulation. This finding may be the result of differences between predominant patterns of shock among various age groups. Neonates and younger children commonly present with right or left ventricular dysfunction. As age increases, the shock becomes predominantly distributive and may be more difficult for VA ECMO to support. Furthermore, the difficulty in achieving adequate flows in older, larger patients could contribute to higher mortality, and should be the subject of further study.

In our study, platelet count of less than 30 000 at the time of cannulation was a predictor of mortality. Thrombocytopenia is a strong independent predictor of disease severity and death in adult ICU patients, and plasmapheresis to treat thrombocytopenia-associated multiorgan failure (TAMOF) has been shown to improve survival in pediatric septic shock patients on ECMO. This finding may be the result of differences between the predominant patterns of shock among the various age groups.^{19–27} Unlike the age and weight cutoffs, the platelet count finding is actionable, offering a potential therapeutic option for these critically ill patients.

Our study is the largest case series to date of pediatric ECMO patients with refractory septic shock exclusively supported with peripheral VA ECMO. We find that adequate ECMO flow rates could be achieved through peripheral cannulation to support patients with septic shock, with none of the mortalities directly attributed to inadequate ECMO flows. Our results reinforce the use of peripheral VA ECMO cannulation as an effective and less invasive method of supporting pediatric patients with septic shock.

Study limitations include its retrospective nature, lack of randomization between peripheral and central ECMO use, and the two-center design, which increases the risk of variability and biases. Over the 15-year study period, changes in practice, technology, and experience were difficult to account for, as were differences in institutional candidacy protocols, ECMO initiation time, cannulation strategies, data collection strategies, and ECMO management between institutions. Additionally, lactate clearance was not evaluated in our study, due to the inconsistent use of lactate as a marker of end-organ perfusion across all patients at both institutions. Another limitation was lack of data on detailed microbiology including causative organisms. To address these limitations, future prospective, multicenter, observational studies of this patient population are required.

Conclusion

This is the largest case series of pediatric ECMO patients with refractory septic shock solely supported with peripheral VA ECMO. Survival was comparable to patients supported by ECMO for other non-cardiac diagnoses. ECMO candidacy for oncology patients should be more thoughtful, deliberate, and include a multidisciplinary team due to high mortality.

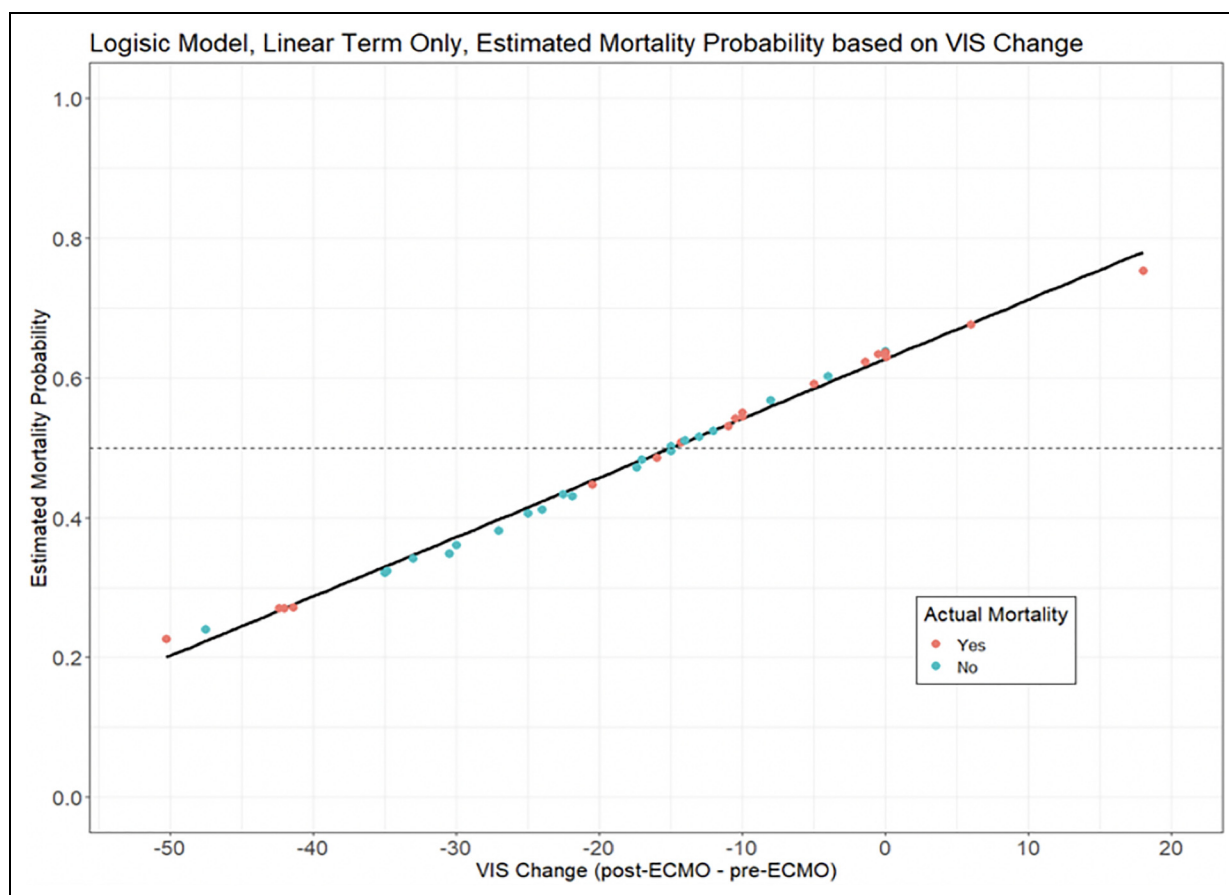


Figure 3. Association between VIS change and predicted mortality probability.

Table 3. Potential Predictors for ECMO Duration.

Independent variable	Incidence rate ratio [95%CI]	P-Value
Age at ECMO *	1.04 [0.74, 1.47]	.802
Admission weight *	1.05 [0.72, 1.53]	.810
VIS *	0.98 [0.68, 1.41]	1.000
Oncology patient	0.81 [0.4, 1.77]	.550
Bacterial growth	0.52 [0.27, 0.96]	.040
Fungal growth	0.71 [0.19, 2.94]	.597
Viral growth	4.69 [2.25, 10.6]	<.001
Blood infection	1.55 [0.81, 2.98]	.192
Lung infection	1.72 [0.94, 3.2]	.076
Urine infection	1.06 [0.44, 2.7]	.916
Worst pH pre-ECMO *	1.26 [0.86, 1.84]	.230
PTT pre-ECMO *	0.92 [0.65, 1.3]	.608
INR pre-ECMO *	0.87 [0.62, 1.3]	.408
Platelets pre-ECMO *	1.15 [0.79, 1.76]	.500
Pre-ECMO ECHO Problems	0.75 [0.39, 1.4]	.358
Pump flow rate, 1 hour *	0.88 [0.60, 1.27]	.472
Vasopressor number	1.33 [0.94, 1.89]	.106

Notes: Variables with a * after them have been converted to Z-scores; they have been scaled so that their mean is 0, and their standard deviation is 1.

Bold denotes statistical significance.

Abbreviations: ECMO, extracorporeal membrane oxygenation; VIS, vasoactive inotropic score; PTT, partial thromboplastin time; INR, international normalized ratio; ECHO, echocardiogram.

Improvement in the VIS score over the first 24 hours of ECMO support may be a useful predictor of survival in these patients. We conclude that peripheral VA ECMO is a safe and effective modality to support pediatric patients with refractory septic shock, provided there is establishment of high ECMO pump flows in the first few hours after cannulation.

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Declaration of Conflicting Interests

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